MLR Institute of Technology

Laxman Reddy Avenue, Dundigal, Quthbullapur (M), Hyderabad – 500 043
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An

Industrial Oriented Mini Project Report

On

"AIR QUALITY INDEX (AQI) PREDICTION & POLLUTION HOTSPOT DETECTION"

Submitted in partial fulfilment of the Requirements
for the award of the degree of

Bachelor of Technology

In

Computer Science & Engineering

(Data Science)

By

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Department of Computer Science & Engineering (Data Science)

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CERTIFICATE

This is to certify that the project entitled **"AIR QUALITY INDEX (AQI) PREDICTION & POLLUTION HOTSPOT DETECTION"** has been submitted by **B. SINDHUJA (23R21A67D8), K. SHRILEKHA (23R21A67F7), V. SAI GANESH (23R21A67K0), and D.LOKESH (23R21A67E2)** in partial fulfilment of the requirements for the award of degree of Bachelor of Technology in Computer Science and Engineering (Data Science) from MLR Institute of Technology affiliated to Jawaharlal Nehru Technological University, Hyderabad. The results embodied in this project have not been submitted to any other University or Institution for the award of any degree or diploma.

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DECLARATION

We hereby declare that the project entitled **"AIR QUALITY INDEX (AQI) PREDICTION & POLLUTION HOTSPOT DETECTION"** is the work done during the period from **July 2025 to March 2026** and is submitted in partial fulfilment of the requirements for the award of degree of Bachelor of Technology in Computer Science and Engineering (Data Science) from MLR Institute of Technology affiliated to Jawaharlal Nehru Technological University, Hyderabad. The results embodied in this project have not been submitted to any other University or Institution for the award of any degree or diploma.

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ABSTRACT

Air pollution is a serious environmental problem that significantly affects human health, climate, and overall quality of life. This project presents an intelligent system for Air Quality Index (AQI) prediction using a combination of IoT sensors and machine learning techniques. Environmental parameters such as gas concentrations, temperature, humidity, and dust particles are continuously collected using sensors connected to a NodeMCU ESP8266, which acts as a microcontroller for data acquisition and transmission. The collected data is sent to a computer system where it undergoes preprocessing steps such as cleaning, normalization, and feature selection using Python. A Random Forest machine learning algorithm is then applied to analyze the data, identify patterns, and accurately predict AQI values based on real-time and historical inputs. The system further classifies air quality into categories such as Good, Moderate, Poor, and Hazardous, enabling users to easily understand pollution levels and take necessary precautions like limiting outdoor activities or using protective measures. Additionally, the system supports real-time monitoring and can be extended to detect pollution hotspots and visualize air quality trends over time. This solution is cost-effective, scalable, and highly useful for applications in smart city development, environmental monitoring, public health awareness, and decision-making by authorities and researchers.

Keywords: Air Quality Index (AQI) Prediction, Pollution Monitoring, IoT-Based System, NodeMCU ESP8266, Environmental Sensors, Random Forest Algorithm, Machine Learning, Real-Time Data Analysis, Pollution Hotspot Detection, Smart City Applications, Environmental Monitoring System

SDG Goals Mapping:

SDG 3: Good Health and Well-being

- Helps monitor air pollution levels that directly impact human health
- Provides AQI predictions to warn people about harmful conditions
- Enables preventive actions like avoiding outdoor exposure or using masks
- Reduces risks of respiratory and cardiovascular diseases

SDG 11: Sustainable Cities and Communities

- Supports smart city development through real-time air quality monitoring
- Helps authorities identify highly polluted areas (hotspots)
- Assists in urban planning and pollution control strategies
- Improves overall environmental quality in cities

SDG 13: Climate Action

- Tracks pollution trends contributing to climate change awareness
- Encourages data-driven actions to reduce emissions
- Helps in understanding environmental patterns for better decision-making
- Promotes proactive measures to control pollution levels

SDG 12: Responsible Consumption and Production

- Creates awareness about pollution caused by human activities
- Encourages responsible use of resources and reduced emissions
- Helps industries and individuals understand their environmental impact
- Promotes sustainable practices through data insights

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CONTENTS

Name of the Topic	Page. No
Certificate	**ii**
Declaration	**iii**
Acknowledgement	**iv**
Abstract	**v**
Chapter 1: INTRODUCTION	
1.1 Overview	**1**
1.2 Objectives	**2**
1.3 Purpose	**2**
1.4 Motivation	**3**
Chapter 2: LITERATURE SURVEY	**3**
2.1 ML-Based Air Pollution Forecasting	**4**
2.1.1 Methodology	4
2.1.2 Advantages	4
2.1.3 Disadvantages	5
2.2 IoT-Based AQI Monitoring System	**5**
2.2.1 Methodology	5
2.2.2 Advantages	6
2.2.3 Disadvantages	6
2.3 AQI Prediction Using Machine Learning	**6**
2.3.1 Methodology	7
2.3.2 Advantages	7
2.3.3 Disadvantages	7
2.4 AI-Based Real-Time Air Monitoring	**7**
2.4.1 Methodology	8
2.4.2 Advantages	8
2.4.3 Disadvantages	8

2.5 Air Quality Prediction Models **9**
2.5.1 Methodology 9
2.5.2 Advantages 9
2.5.3 Disadvantages 10
2.6 Summary Table **11**
Chapter 3: PROPOSED SYSTEM **12**
3.1 System Architecture **12**
3.2 Module Division
3.2.1 Sensor Data Acquisition Module 15
3.2.1.1. Description
3.2.2 Data Transmission Module 15
3.2.2.1. Description
3.2.3 Data Preprocessing Module 15
3.2.3.1. Description
3.2.4 AQI Prediction Module 15
3.2.4.1. Description
3.2.5 Air Quality Classification Module 16
3.2.5.1. Description
3.2.6 Visualization and Alert Module 16
3.2.6.1. Description
3.2.7 Pollution Hotspot Detection Module 16
3.2.7.1. Description
3.3 UML Diagrams **17**
3.3.1. Use case Diagram 17
3.3.2 Class Diagram 17
3.3.3 Sequence Diagram 18
3.4 Requirements **18**
3.4.1 Software Requirements 18
3.4.2 Hardware Requirements 19

Chapter 4: IMPLEMENTATION **20**

4.1 Methodologies 20

4.2 Technologies Used 21

4.3 Source Code 23

Chapter 5: RESULTS **62**

5.1 Datasets Used 62

5.2 Output Screenshots 63

Chapter 6: CONCLUSION AND FUTURE ENHANCEMENTS **65**

6.1 Conclusion **65**

6.2 Future Enhancements **65**

References **67**

List of Figures:

S. No	Figure No	Name of the Figure
1	3.1	Architecture Workflow Diagram of AQI System
2	3.3.1.1	Use Case Diagram for AQI Prediction System
3	3.3.2.1	Class Diagram of AQI Monitoring System
4	3.3.3.1	Sequence Diagram of Data Flow
5	5.2.1	NodeMCU ESP8266 with Sensor Connections
6	5.2.2	IoT-Based Air Quality Monitoring Block Diagram
7	5.2.3	Data Preprocessing Workflow
8	5.2.4	Random Forest Model Architecture
9	5.2.5	AQI Prediction Output Graph

List of Tables:

S. No	Table No.	Name of the Table
1	2.6	Literature Survey IEEE Summary Table